

Translated

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Primitive Root

Definition

In modular arithmetic, a number g is called a **primitive root modulo n** if every number coprime to n is congruent to a power of g modulo n . Mathematically, g is a **primitive root modulo n** if and only if for any integer a such that $\gcd(a, n) = 1$, there exists an integer k such that:

$$g^k \equiv a \pmod{n}.$$

g is then called the **index** or **discrete logarithm** of a to the base g modulo n . g is also called the **generator** of the multiplicative group of integers modulo n .

In particular, for the case where n is a prime, the powers of primitive root runs through all numbers from 1 to $n-1$.

Existence

Primitive root modulo n exists if and only if:

- n is $1, 2, 4$, or
- n is power of an odd prime number $(n = p^k)$, or
- n is twice power of an odd prime number $(n = 2 \cdot p^k)$.

This theorem was proved by Gauss in 1801.

Relation with the Euler function

Let g be a primitive root modulo n . Then we can show that the smallest number k for which $g^k \equiv 1 \pmod{n}$ is equal to $\phi(n)$. Moreover, the reverse is also true, and this fact will be used in this article to find a primitive root.

Furthermore, the number of primitive roots modulo n , if there are any, is equal to $\phi(\phi(n))$.